A Multi-Modal Accessibility Analysis of Essential Services in Shenzhen

1. Introduction

For several decades, the urban development model are centered on automobiles, which has quietly transformed the focus of urban transportation development towards a predominantly vehicle-oriented transportation system, even leading to significant imbalances in public transportation. In cities, the excessive commuting time for those individuals that do not own private cars and the uneven distribution of available transportation services have placed an disproportionate burden on this group. This has prompted calls for achieving more equitable transportation equity in urban planning (Littman, 2002; Martens, 2017). As a result, modern concepts such as "15-minute cities" have gained popularity, aiming to ensure that residents can meet their daily needs within walking, cycling, or public transportation distances (Moreno et al., 2021). These concepts emphasize the proximity of essential urban facilities to ensure that different social groups do not experience disparities in access to basic public facilities based on their residential locations. By improving the accessibility of resources such as schools, medical facilities, and parks at the community level, reliance on private cars is reduced. As a result, it can enhance the livability and sustainability of the city. This planning concept, based on the pursuit of the nearest transportation facilities, matches the goal of fairly distributing travel convenience and opportunities among different groups, and has gradually become a key indicator for evaluating these fairness outcomes in transportation planning.

However, for some large cities with extremely rapid urbanization progress, achieving accessibility equity remains a challenge. For instance, in China's megacities, although the new subway systems have improved the overall public transportation accessibility, they have also widened the accessibility gap between different regions, resulting in the lack of some basic services around some remote communities. Compared with walking, cycling, or other modes of transportation, private cars usually save a significant amount of time and can benefit from the advantages of some express roads and public facilities guidance, which makes people who do not drive face restrictions such as detours or longer travel times to reach certain specific areas. Therefore, using simulation experiments to compare the accessibility of various transportation methods is crucial for revealing these hidden inequalities. This method can more realistically highlight the social impact of traffic inequality and more clearly demonstrate the disadvantages experienced by those residents who cannot choose faster travel methods (Golub and Martens, 2014; Kawamoto and Shen, 2007).

Based on this background, this study simulated the multimodal accessibility of the multi-area including Shenzhen Bay - Qianhai - Hailai in Shenzhen, China, to provide relevant evidence for the unfairness of public facility accessibility. The study utilized the ArcGIS Pro network analysis tool to assess the travel time from 77 residential hexagonal grid points to important service facilities (hospitals, schools and parks). The travel modes included driving, subway, walking and cycling. This analysis quantified the mode privilege gap within the experimental area, to some extent, proving that car users enjoy faster travel speeds than pedestrians, cyclists or public transportation passengers. By mapping and evaluating these differences, the study provided certain spatial evidence to support urban planning intervention measures. The research results offer insights into how to improve the accessibility of all modes through integrated land use and transportation strategies, and help achieve a more equitable urban transportation system and a more reasonable public facility service setup.

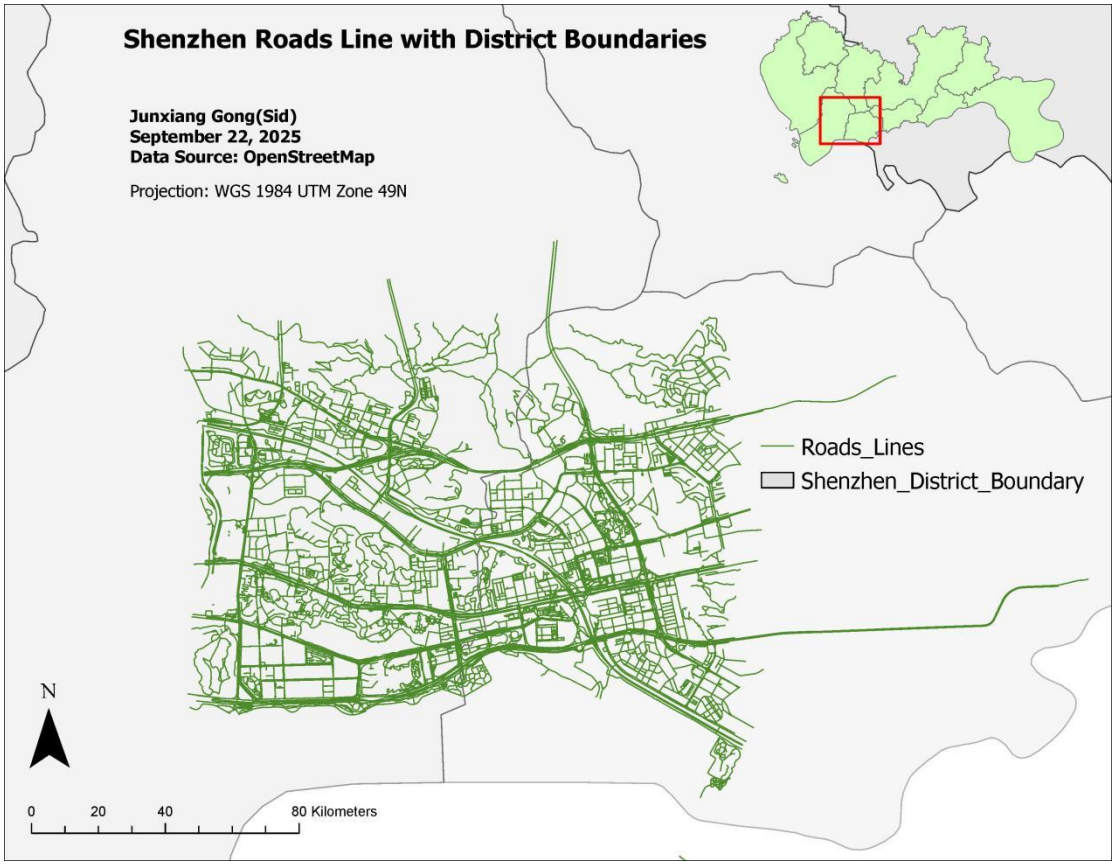
2. Method and data

2.1 Study area

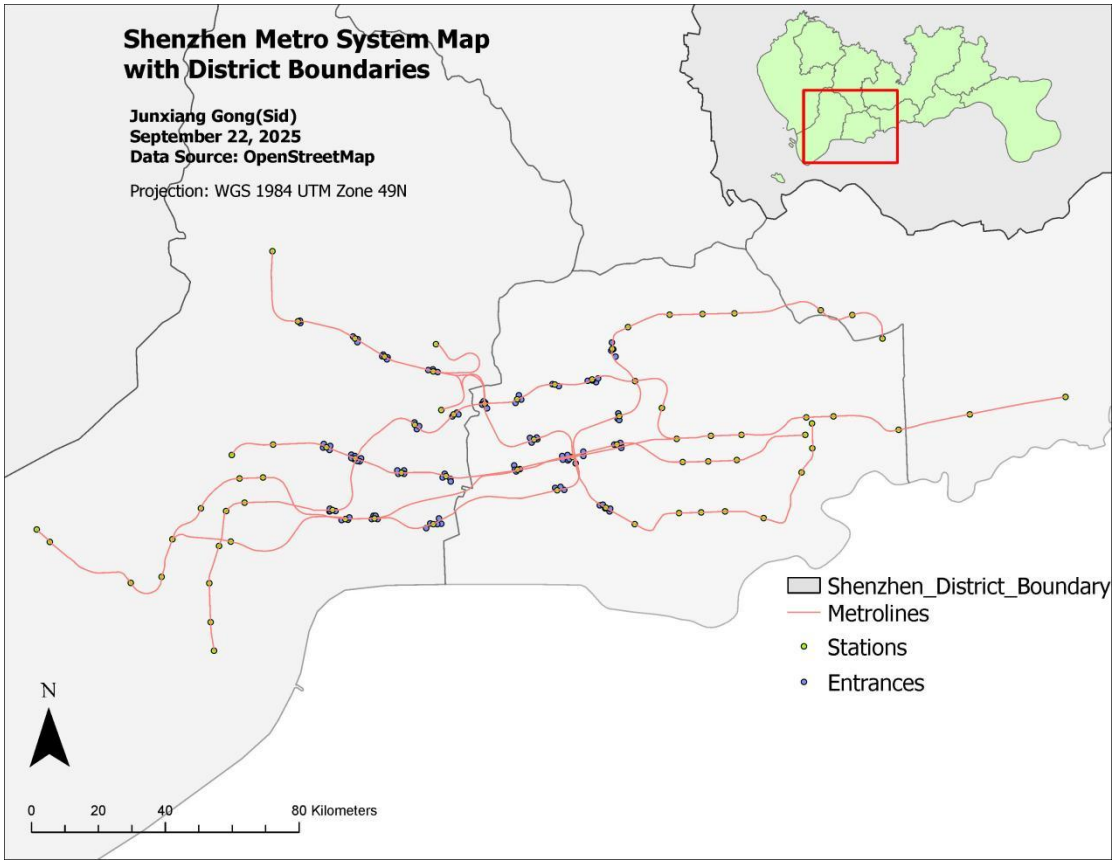
This study selected the Shenzhen Bay - Qianhai - Hailai Corridor area in Shenzhen as the experimental region to examine the differences in traffic accessibility among various transportation modes. The selected area spans the Nanshan District in the west of Shenzhen, extends along Shenzhen Bay, and covers new developed areas like Linhai and Honghai Headquarters District. Statistically, the study area covers the range of 113.955°E to 114.035°E, and 22.495°N to 22.575°N, corresponding to an area of approximately 73 square kilometers. This area is not only the focus of urban investment and policy innovation in Shenzhen in recent decades, but also a pilot area for Shenzhen's public transportation-oriented development and regional integration (Shenzhen Coastal Administration, 2016). There are a large number of infrastructure projects in this area, including new subway lines and intercity railway connection lines. At the same time, there are mixed high-end development projects and aging communities in this area, which to a certain extent can represent the different travel choices of people of different incomes and ages, and also provide a research basis for proving the modern transportation network and the insufficient services (Wang, 2025). Therefore, these characteristics make the Shenzhen Bay - Qianhai - Hailai Corridor area a representative research area for studying urban transportation equity.

In order to comprehensively analyze the accessibility of different transportation modes to the facilities within a specific area, the study area was divided into a grid composed of multiple equal hexagonal units, which covered all the major residential areas within the experimental scope. In this study, the center points of each hexagon were selected based on the actual residential situation and satellite maps, and the points that matched the real situation were taken as the analysis locations. A total of 77 valid residential points were selected. Using the hexagonal grid can provide consistent spatial resolution and minimize boundary and direction deviations, enabling more precise simulation of real conditions. (Birch et al., 2007). At the same time, by using the hexagonal grid and selecting 77 units based on the actual situation, this experiment can generate a smoother spatial pattern and make the distances from the center of the units to the surrounding points more uniform, which is very beneficial for evaluating travel time and connectivity.

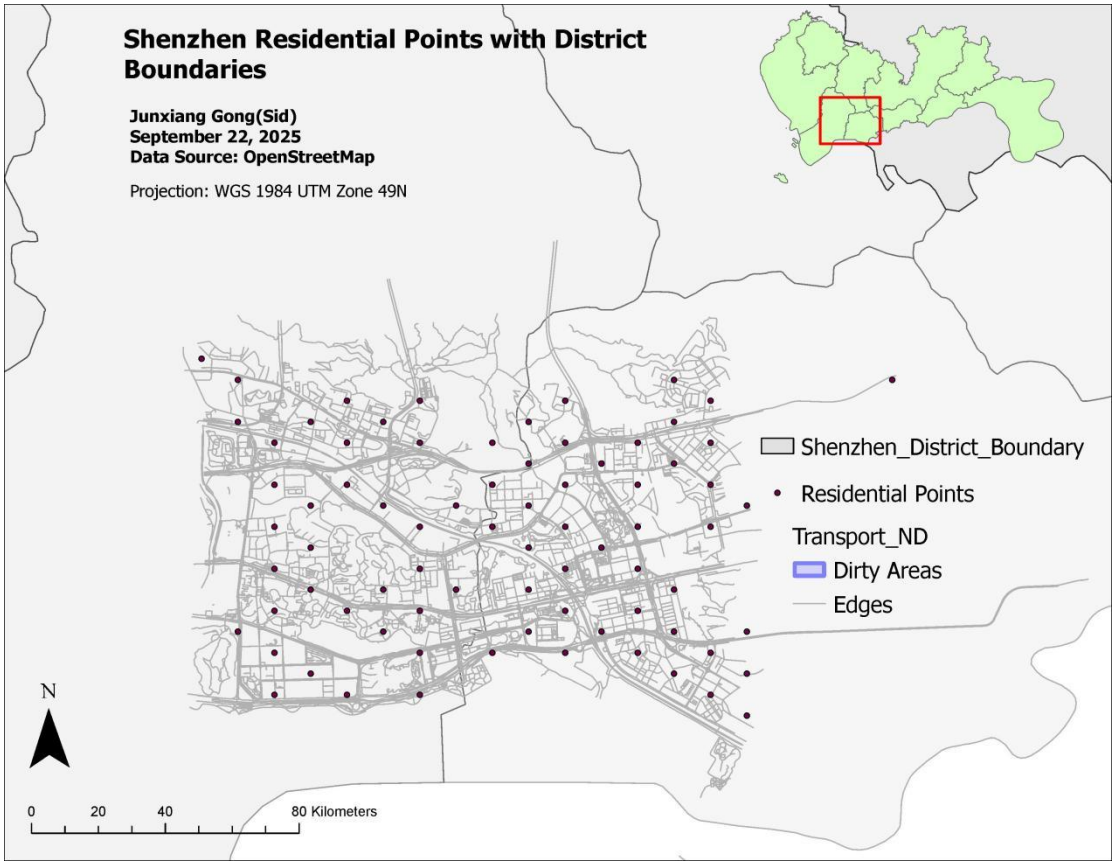
Map 1



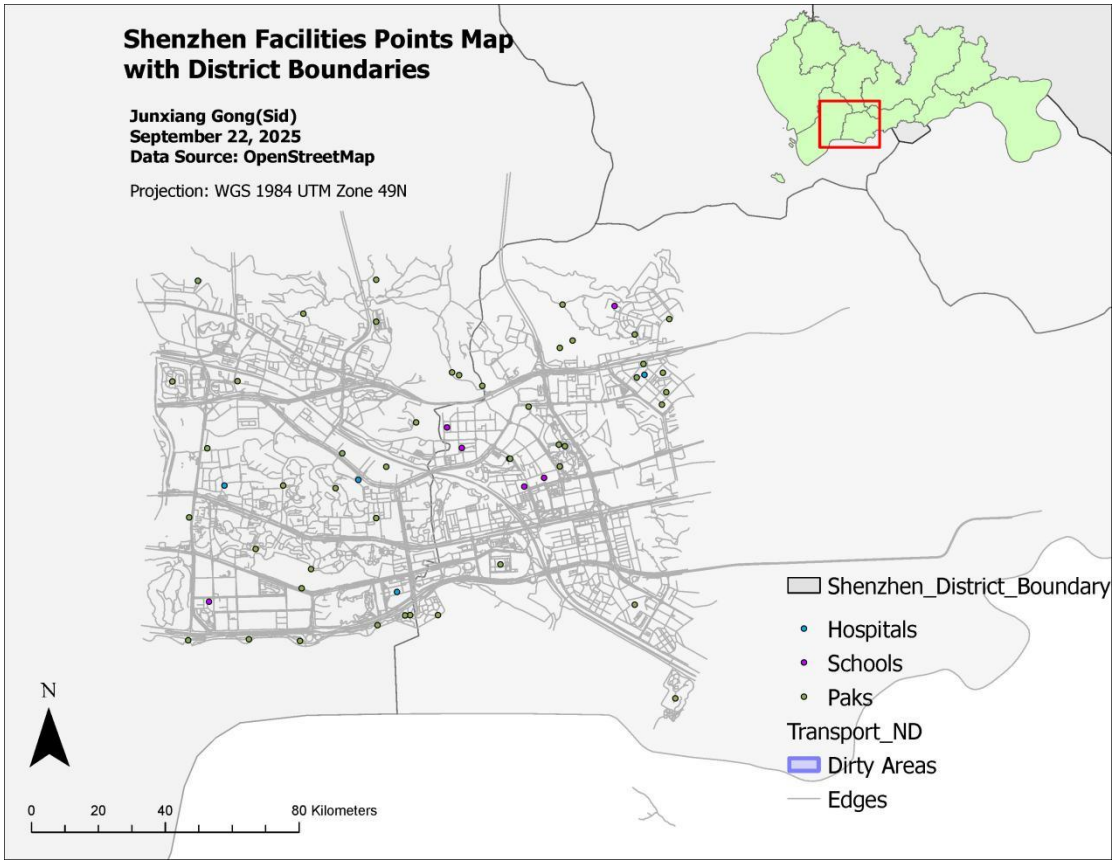
Map 2


Shenzhen Metro System Map
with District Boundaries
Junxiang Gong(Sid)
September 22, 2025
Data Source: OpenStreetMap
Projection: WGS 1984 UTM Zone 49N
Shenzhen_District_Boundary
Metrolines
Stations
Entrances
N
0
20
40
80 Kilometers

Map 3


Shenzhen Residential Points with District
Boundaries
Junxiang Gong(Sid)
September 22, 2025
Data Source: OpenStreetMap
Projection: WGS 1984 UTM Zone 49N
Shenzhen_District_Boundary
Residential Points
Transport_ND
Dirty Areas
Edges
N
0
20
40
80 Kilometers

Map 4



2.2 Data and Parameters

The road network data in this paper, including highways, main roads, local streets and sidewalks, was extracted from the public data of OpenStreetMap using a self-written Python program and apply an API. The data was collected in November 2025. At the same time, the subway system data was also retrieved using the same method, including subway lines, stations and entrances, as well as points of interest such as hospitals, schools and parks. After obtaining all the data, the real satellite images were used to manually verify each service point and stations against the official list of 2025 to ensure its accuracy. Then, all the spatial data were projected and integrated in ArcGIS Pro to form the basis for calculating the travel time of the network as described below. In short, within the study area, the number of relevant facilities is 4 hospitals, 6 schools, and 46 parks. Meanwhile, the number of effective stations points and entrances points is 75 and 253 respectively.

Using the network analysis tools of ArcGIS Pro, this paper respectively constructed travel network models based on highways and subways. In this study, the road dataset was filtered, only the drivable and walkable sections were retained, and data cleaning

was carried out to ensure the correct topology of intersections. Based on that, this paper sets three modes of travel on road network: driving, cycling, and walking. Since the focus of the analysis was on travel time rather than vertical separation, multi-level structures such as elevated bridges and tunnels were simplified to a single planar network. At the same time, to ensure the feasibility of the experiment, one-way restrictions were not modeled, congestion effects were not included, all turns were given a certain time punishment, and all intersection situations were covered. To reflect the actual travel speeds and conditions, each road segment is assigned a speed specific to the mode. For driving, we adopted the average peak-hour speed from local traffic data and set it at 35 kilometers per hour. According to relevant research, the speed of expressways is approximately 75 kilometers per hour, main arterial roads are approximately 45 kilometers per hour, urban roads are approximately 30 kilometers per hour, and small streets are approximately 20 kilometers per hour (Tian et al., 2022). The speed of bicycles on accessible roads is approximately 12 kilometers per hour, and the speed of pedestrians on sidewalks is 4.8 kilometers per hour (Tian et al., 2022).

For subway network, the subway original data requires additional processing because raw OpenStreetMap subway lines and stations in this area do not always align perfectly with official route maps and high resolution imagery. Therefore, in this research, the subway lines were redrawn as single two dimensional polylines that follow the actual alignment of each route and station points were shifted to match visible entrances and platforms. Besides, this paper also set an appropriate speed parameters for subway as 33.3 kilometers per hour.

3. Method

3.1 Network

In this paper, for the modes of driving, walking, cycling, and traveling, the travel times are calculated based on the network impedance. This impedance is converted from the link length to minutes by transforming it according to a specific speed pattern. Under these three travel modes, all ground travel calculations use the road network dataset. Each road segment stores its length in meters and three impedance fields in minutes. These impedance fields represent walking time, cycling time, and driving time respectively. They are calculated by applying a constant running speed to the same segment length. Specifically, the walking speed is set at 4.8 kilometers per hour. The cycling speed is set at 12.0 kilometers per hour. The driving speed is set at the effective average speed of 35.0 kilometers per hour. Through the traffic network

analysis method of the Closest Facilities, the solver accumulates the minutes at the segment level along the candidate routes and returns the path with the shortest total travel time.

For subway, its travel time was computed using a chained structure,by utilizing Closest Facilities across two separate networks. Meanwhile, for the time calculation of the subway section in this experiment, a three-part combined calculation method is adopted. Specifically, the entire subway travel time from the starting point to the facility is equal to the travel time from the starting point to the entrance of the selected station, plus the inter-station time from the starting point of the subway to the destination point, and then plus the travel time from the entrance of the destination station to the entrance of the facility. The parts of entering the underground from the ground and going from the underground to the nearest facility are calculated based on the walking resistance in the road network. The station-to-station part is calculated based on the constant subway running speed (33.3 kilometers per hour) in the subway network. Finally, this experiment matches the three output tables generated in the workflow to ensure that the subway travel time is represented as a door-to-door travel time, and to guarantee that the selection of each subway line is the shortest time. Besides, in order to facilitate the calculation of the subway transportation network, this experiment adopts stations as the time calculation points, rather than entrances.

3.2 Measures

Equity in this study is evaluated through an accessibility based perspective, where fairness is assessed by comparing how quickly residents can reach essential services under different travel modes. This framing follows the view that accessibility, rather than mobility alone, is the relevant outcome for distributive questions in transport planning (Martens, 2016) and city equity verification(Pereira, Schwanen, & Banister, 2017). Travel time is used as the primary accessibility cost because it is a standard and interpretable impedance in location based accessibility measurement (Geurs & van Wee, 2004).

For each residential origin i, and for each facility category k including hospitals, schools, and parks, the analysis first computes a minimum travel time $T_m(i,k)$under each mode m. The value $T_m(i,k)$ represents the shortest travel time from origin iii to the nearest facility in category k under mode m. This minimum time is produced by network based shortest path solutions and nearest facility selection, which enables consistent comparison across modes when the same spatial units and facility sets are used (Geurs & van Wee, 2004). Driving is treated as the reference mode because it

reflects the fastest option in many urban contexts and serves as a baseline against which transport disadvantage can be expressed (Kawabata & Shen, 2007).
The core equity indicator is the modal privilege gap. It operationalizes the time advantage of private car travel relative to a noncar mode. For each origin i, facility category$k$, and comparison mode m in subway, bike, and walk, the gap is computed as

$$\mathrm{Gap}_m(i,k)=T_m(i,k)-T_{drive}(i,k)$$

This structure is consistent with the modal accessibility gap concept, which uses differences between modal accessibility outcomes to reveal imbalance in transport systems (Kwok & Yeh, 2004). A positive gap indicates that the comparison mode is slower than driving at origin $i$for category k, which is interpreted as car privilege at that location. A gap equal to zero indicates parity. A negative gap indicates that the comparison mode is faster than driving. This sign based interpretation aligns with prior studies that quantify car versus transit disparities using travel time and accessibility differences (Benenson, Martens, Rofé, & Kwartler, 2011) and (Kawabata & Shen, 2007).

Equity outcomes are summarized across all origins in the study area by reporting the share of origins with $\mathrm{Gap}_m(i,k)$ greater than zero, and by reporting the central tendency and spread of $\mathrm{Gap}_m(i,k)$such as mean and median. This produces a distributional view of car privilege rather than a single average, which is consistent with equity oriented evaluation that emphasizes who benefits and where the disadvantages concentrate (Pereira, Schwanen, & Banister, 2017).

4. Result

In the road-based analysis, this paper examines three different facility points and, by applying driving, cycling, and walking travel modes, generates a total of nine route result tables. Specifically, each table represents the shortest travel time from each community starting point to the nearest facility within the specified destination type. Additionally, this experiment aggregates these nine tables into a community-level dataset to ensure that each community can obtain comparable shortest time datasets categorized by travel mode and destination type.

In the analysis based on the subway, this experiment produced three tables for three different facilities respectively. They are the shortest route table from residential areas to stations, the OD table from stations to stations, and the shortest route table from stations to facility points. Specifically, for each destination group, this experiment generated a route table on the road network from communities to stations or entrance

passages. The travel distance between stations on the subway network was generated. At the same time, a second route table was generated for the station-to-facility passages on the road network. By integrating these three tables, the total subway travel time from each community to the nearest facility within its destination group was ultimately obtained.

Overall, private cars have demonstrated the advantage of the shortest travel time in all three types of facilities, and they have also shown certain traffic privileges in all communities. Non-driving modes not only have a higher overall time cost but also exhibit greater variability across communities, which means that they face unfair traffic accessibility while also having uncertainty in traffic outcomes. By facility type, the gap in hospital accessibility is the most prominent. This is because there are fewer and more scattered hospital facilities in this area. This can be reflected to some extent in the condition of relatively sparse facility supply and longer travel distances, where non-driving groups bear greater traffic time pressure and face higher transportation costs. In the school scenario, cycling is relatively more competitive, specifically manifested in its closest performance to driving time, and even at certain points, it can serve as an alternative transportation mode, indicating its potential value as a low-cost improvement strategy. In the park scenario, walking and cycling can still achieve a relatively acceptable time level in some communities, but the subway, due to the fixed costs of entry and exit, shows a certain disadvantage in short-distance travel efficiency, resulting in a wider gap with driving overall.

Shown in figure. 1, figure. 2, figure. 3, the results for different facilities further clarify that this inequality is more pronounced in different destination environments. For parks, the driving speed is still very fast, approximately 1.6 minutes; cycling takes about 4.7 minutes; walking takes about 11.4 minutes; and taking the subway is the slowest, approximately 18.1 minutes, which is consistent with the fact that the total subway travel cost including entry and exit fees is still not low. The gap distribution between parks is significantly skewed to the right, with 55.8% of communities exceeding the 15-minute gap between subway and driving. For schools, driving takes approximately 4.4 minutes, cycling takes approximately 11.9 minutes, taking the subway takes approximately 17.6 minutes, and walking takes approximately 29.8 minutes. This makes cycling the most competitive non-driving mode of transportation. Compared to parks, schools have fewer cases of significant gaps, with 28.6% of communities exceeding the 15-minute threshold. For hospitals, the gap is the most significant, driving takes approximately 4.2 minutes, cycling takes approximately 11.9 minutes, taking the subway takes approximately 22.9 minutes, and walking takes approximately 29.8 minutes. The distribution of hospital service coverage shows that

the severity is the highest, with 70.1% of communities exceeding the 15-minute travel time standard. This indicates that in this study area, reliance on public transportation may cause significant time losses in accessing emergency medical services.

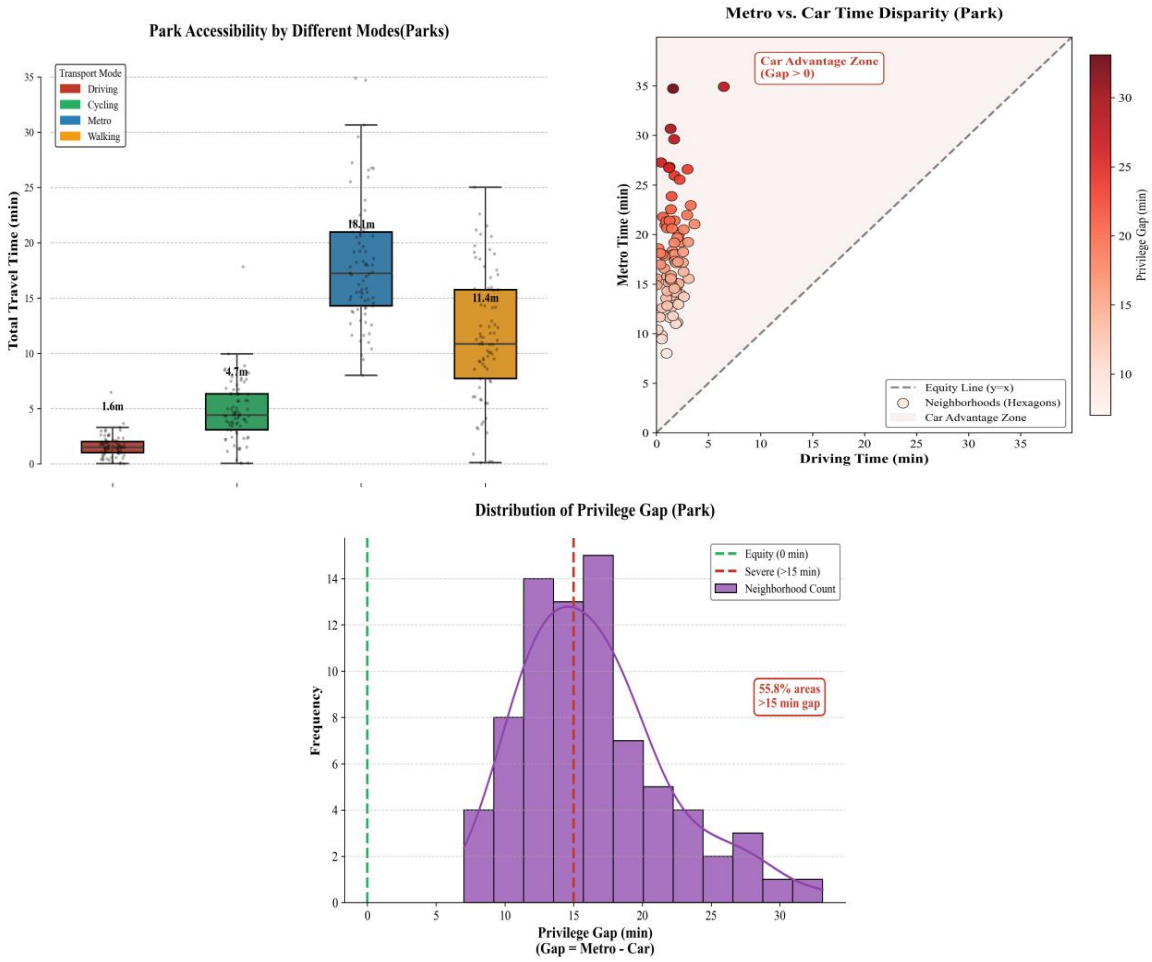


Figure. 1.

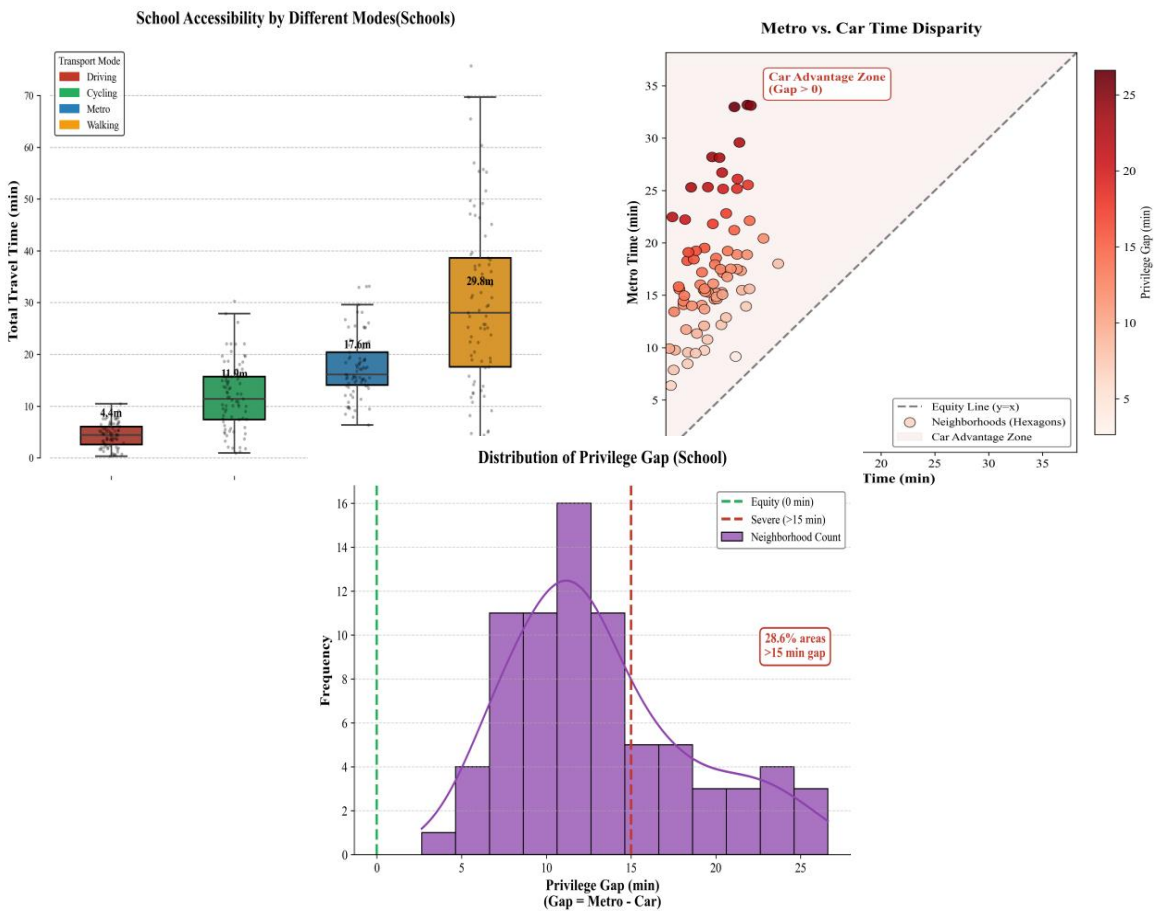

School Accessibility by Different Modes(Schools)
Transport Mode
Driving
Cycling
Metro
Walking
Total Travel Time (min)
4.4m
11.9m
17.5m
29.8m
Metro vs. Car Time Disparity
Car Advantage Zone
(Gap > 0)
Metro Time (min)
Time (min)
Privilege Gap (min)
Equity Line (y=x)
Neighborhoods (Hexagons)
Car Advantage Zone
Distribution of Privilege Gap (School)
Equity (0 min)
Severe (>15 min)
Neighborhood Count
28.6% areas
>15 min gap
Frequency
Privilege Gap (min)
(Gap = Metro - Car)

Figure. 2.

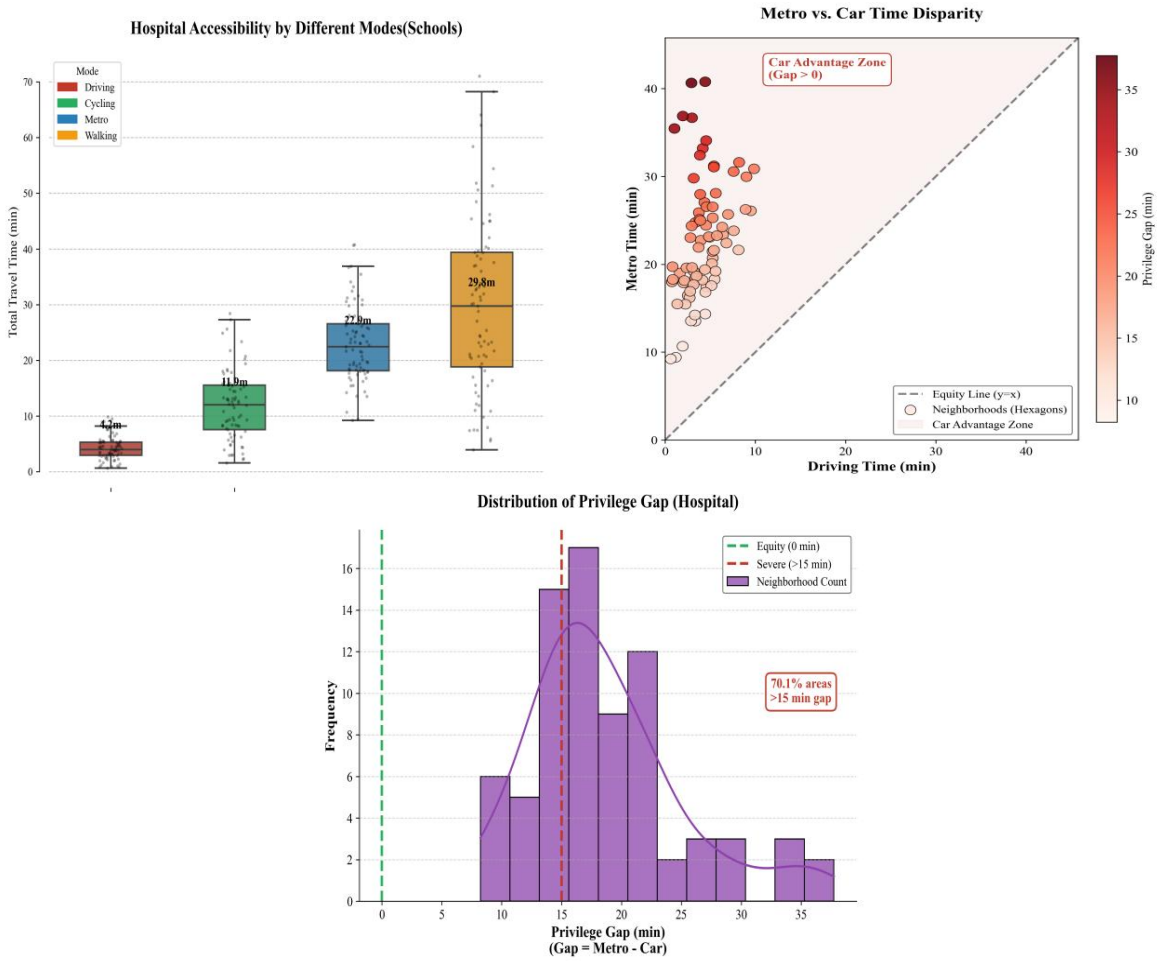


Figure. 3.

5. Conclusion and discussion

This study simulated the actual traffic environment within the area from Shenzhen Bay to Qianhai and Hailai in ArcGIS Pro. Based on hexagonal neighborhood units and an independent yet interconnected road and subway network constructed based on real OpenStreetMap data, along with artificial corrections and door-to-door time calculations, it proved that there exists a clear and stable hierarchical structure in the traffic accessibility of the central area of such large cities. Among them, driving is always the fastest mode of transportation.

Future work for this experiment could expand this framework to include congestion, public transportation networks and income data, and could test whether similar patterns would emerge in other Chinese cities and in low-density urban areas. Even in its current form, this analysis indicates that if planning practices remain centered on the speed of cars, such practices may lead to a normalization of the distribution of

traffic flow, and this distribution would systematically put those without cars at a disadvantage, especially in areas crucial to inclusive urban life such as health, education and daily entertainment.

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